

MECHATRONICS AND IOT

ASSIGNMENT - 9

Design and Develop an Automated Color-Based Product Sorting System for Manufacturing Industries

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Design and Develop an Automated Color-Based Product Sorting System for Manufacturing Industries

1. Project Overview

The Automated Color-Based Product Sorting System is an industrial automation project designed to automatically detect and sort products according to their color. The system reduces manual sorting effort and improves the speed, consistency, and accuracy of product handling in manufacturing industries.

The proposed system uses a **TCS3200 color sensor** to identify the color of objects placed on a conveyor belt. An **Arduino UNO** acts as the main controller and processes the color information received from the sensor. Based on the detected color, a servo motor operates a sorting mechanism and directs the product into the appropriate section.

An IR sensor is used to detect the presence of a product. A DC motor drives the conveyor belt, while an L298N motor driver is used to control the conveyor motor. LEDs can be used to indicate the detected color.

Main components:

- TCS3200 Color Sensor
- Arduino UNO
- IR Sensor
- Servo Motor
- LEDs
- L298N Motor Driver
- Jumper wires.

2. Project Statement

In many manufacturing industries, products must be separated according to their color before packaging, processing, or further production. Manual color-based sorting requires human effort and can result in inconsistent sorting, reduced productivity, and errors.

Therefore, an automated color-based product sorting system is proposed. The system detects the color of products using a color sensor and automatically directs each product to its corresponding category using a motorized sorting mechanism.

The system aims to improve sorting efficiency, reduce human intervention, and demonstrate the application of embedded systems in industrial automation.

3. Proposed Solution

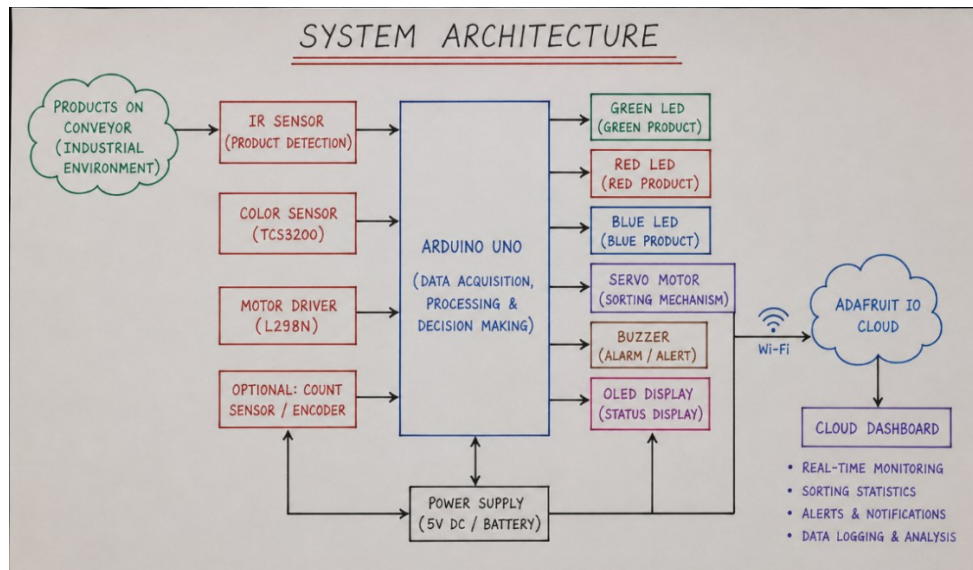
The proposed system consists of a conveyor belt, color sensor, IR sensor, Arduino UNO, motor driver, DC motor, and servo-based sorting mechanism.

The system operates as follows:

- The conveyor belt carries products toward the detection area.
- The IR sensor detects the presence of a product.
- The conveyor temporarily stops or slows down.
- The TCS3200 color sensor detects the color of the product.
- The Arduino processes the sensor output.
- The detected color is compared with predefined color ranges.
- The corresponding LED is activated.
- The servo motor moves to the appropriate sorting position.
- The product is pushed or directed into the corresponding container.

- The servo returns to its initial position.
- The conveyor starts again.
- The process repeats continuously.

4. System Architecture:

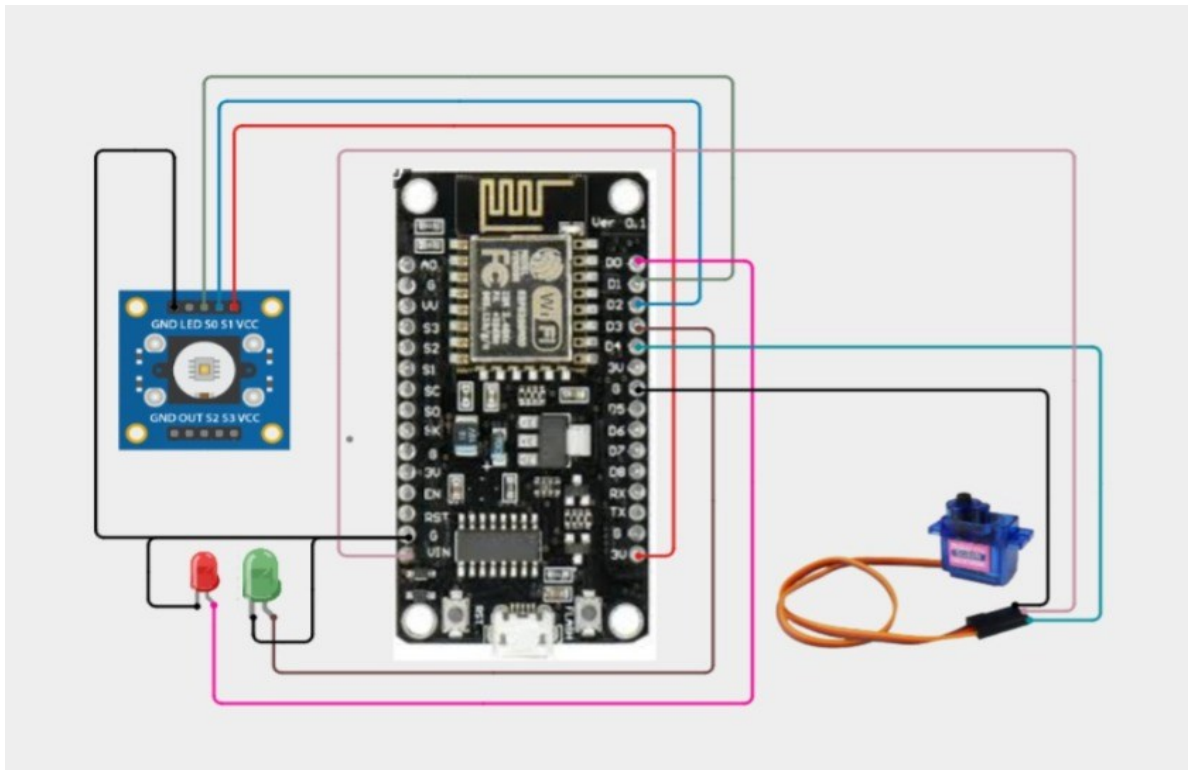


5. Circuit Table:

CIRCUIT TABLE

S. No.	Component	Pin	Arduino UNO Pin	Description
1.	TCS3200 Color Sensor	VCC GND S0 S1 S2 S3 OUT OE	5V GND D4 D5 D6 D7 D8 GND	Power Supply (5V) Ground Frequency Scaling Frequency Scaling Color Filter Selection Color Filter Selection Frequency Output Output Enable (Active LOW)
2.	IR Obstacle Sensor	VCC GND OUT	5V GND D2	Power Supply (5V) Ground Digital Output (Product Detection)
3.	Servo Motor	VCC GND Signal	External 5V GND D9	Power Supply (External 5V) Ground Servo Signal
4.	Red LED	Anode (+) Cathode (-)	D10 (through 220Ω) GND	Red LED (Red Product) Ground
5.	Green LED	Anode (+) Cathode (-)	D11 (through 220Ω) GND	Green LED (Green Product) Ground
6.	Blue LED	Anode (+) Cathode (-)	D12 (through 220Ω) GND	Blue LED (Blue Product) Ground
7.	L298N Motor Driver (To DC Motor)	IN1 IN2 ENA OUT1 / OUT2 GND VMS (Motor Supply)	D3 D13 PWM Pin (e.g. D5) To DC Motor GND External Supply (6V - 12V)	Motor Direction Motor Direction Motor Speed (PWM) Motor Connections Common Ground Motor Power Supply
8.	Power Supply	5V GND	5V to System GND	System Power Supply Ground

6. Circuit Design:



7. Algorithm:

Step-by-step algorithm:

1. Start the system.
2. Initialize Arduino pins.
3. Initialize the color sensor.
4. Initialize the IR sensor.
5. Initialize the servo motor.
6. Start the conveyor motor.
7. Continuously monitor the IR sensor.
8. If no product is detected, continue the conveyor.
9. If a product is detected, stop the conveyor.
10. Read the color sensor output.

- 11.Determine the dominant color.
- 12.Compare the detected color with predefined values.
- 13.Turn ON the corresponding LED.
- 14.Move the servo to the appropriate sorting position.
- 15.Allow the product to enter the corresponding sorting bin.
- 16.Return the servo to its initial position.
- 17.Start the conveyor again.
- 18.Repeat the process.

8. Coding:

```
#include <ESP8266WiFi.h>

#include "AdafruitIO_WiFi.h"

#include <Servo.h>

// =====
// Wi-Fi DETAILS
// =====

#define WIFI_SSID   "OnePlus Nord CE5 7C5C"
#define WIFI_PASSWORD "qkts5737"
// =====

// ADAFRUIT IO DETAILS
// =====

#define IO_USERNAME  "Sivasaravanan"
#define IO_KEY       "aio_ADVS50c7oG5Sx1R3FskFAq0fjUrB"
// Create Adafruit IO connection
AdafruitIO_WiFi io(IO_USERNAME, IO_KEY, WIFI_SSID, WIFI_PASSWORD);

// =====
// TCS3200 CONNECTIONS
// =====
```

```
#define S0 D1

#define S1 D2

#define S2 D5

#define S3 D6

#define SENSOR_OUT D

// =====

// SERVO

// =====

#define SERVO_PIN D4

Servo sorterServo

// =====

// LEDS

// =====

#define LED1 D0

#define LED2 D3

// =====

// ADAFRUIT IO FEEDS

// =====

AdafruitIO_Feed *colorFeed = io.feed("color");

AdafruitIO_Feed *servoFeed = io.feed("servo");

AdafruitIO_Feed *redFeed = io.feed("red");

AdafruitIO_Feed *greenFeed = io.feed("green");

AdafruitIO_Feed *blueFeed = io.feed("blue");

// =====

// READ RED

// =====

int readRed()

{

    digitalWrite(S2, LOW);
```

```
digitalWrite(S3, LOW);

delay(100);

int value = pulseIn(SENSOR_OUT, LOW, 100000);

return value;

// =====
// READ GREEN
// =====

int readGreen()
{
    digitalWrite(S2, HIGH);
    digitalWrite(S3, HIGH)
    delay(100)
    int value = pulseIn(SENSOR_OUT, LOW, 100000);

    return value;
}

// =====
// READ BLUE
// =====

int readBlue()
{
    digitalWrite(S2, LOW);
    digitalWrite(S3, HIGH);
    delay(100);
    int value = pulseIn(SENSOR_OUT, LOW, 100000);

    return value;
}

// =====
// SETUP
```



```

// =====
void setup()
{
  Serial.begin(115200);
  // ----- TCS3200 -----
  pinMode(S0, OUTPUT);
  pinMode(S1, OUTPUT);
  pinMode(S2, OUTPUT);
  pinMode(S3, OUTPUT);
  pinMode(SENSOR_OUT, INPUT);
  // 20% frequency scaling
  digitalWrite(S0, HIGH);
  digitalWrite(S1, LOW);
  // ----- LEDs -----
  pinMode(LED1, OUTPUT);
  pinMode(LED2, OUTPUT);
  digitalWrite(LED1, LOW);
  digitalWrite(LED2, LOW);
  // ----- Servo -----
  sorterServo.attach(SERVO_PIN);
  // Start position
  sorterServo.write(90);
  // ----- Adafruit IO -----
  Serial.println();
  Serial.println("=====");
  Serial.println("COLOR SORTING SYSTEM");
  Serial.println("=====")
  Serial.println("Connecting to Adafruit IO...");
  io.connect();

```

```

while (io.status() < AIO_CONNECTED)
{
    Serial.print(".");
    delay(500);
}

Serial.println();

Serial.println("Adafruit IO Connected!");

Serial.println();
}

// =====

// LOOP

// =====

void loop()
{
    // Keep Adafruit IO connected
    io.run();

    // =====

    // READ COLOR VALUES

    // =====

    int redValue = readRed();
    int greenValue = readGreen();
    int blueValue = readBlue();

    // =====

    // SERIAL MONITOR

    // =====

    Serial.println("-----");
    Serial.print("RED  = ");
    Serial.println(redValue);
    Serial.print("GREEN = ");

```

```
Serial.println(greenValue);

Serial.print("BLUE = ");

Serial.println(blueValue);

// =====
// SEND RGB VALUES TO ADAFRUIT
// =====

redFeed->save(redValue);

greenFeed->save(greenValue);

blueFeed->save(blueValue);

// Turn both LEDs OFF

digitalWrite(LED1, LOW);

digitalWrite(LED2, LOW);

// =====
// RED DETECTION
// =====

if (redValue < greenValue &&
    redValue < blueValue &&
    redValue > 0)
{
    Serial.println("COLOR DETECTED = RED");

    // LED 1 ON

    digitalWrite(LED1, HIGH);

    // Servo position

    sorterServo.write(30);

    // Send data to Adafruit

    colorFeed->save("RED");

    servoFeed->save(30);

    delay(1000);
}
```

```

// =====
// GREEN DETECTION
// =====
else if (greenValue < redValue &&
        greenValue < blueValue &&
        greenValue > 0)
{
    Serial.println("COLOR DETECTED = GREEN");
    // LED 2 ON
    digitalWrite(LED2, HIGH)
    // Servo position
    sorterServo.write(90)
    // Send data to Adafruit
    colorFeed->save("GREEN");
    servoFeed->save(90);
    delay(1000);
}

// =====
// BLUE DETECTION
// =====
else if (blueValue < redValue &&
        blueValue < greenValue &&
        blueValue > 0)
{
    Serial.println("COLOR DETECTED = BLUE");
    // Servo position
    sorterServo.write(150);
    // Send data to Adafruit
    colorFeed->save("BLUE");

```

```

servoFeed->save(150)

delay(1000);

}

// =====
// UNKNOWN COLOR
// =====

else

{

  Serial.println("COLOR DETECTED = UNKNOWN")

  sorterServo.write(90);

  colorFeed->save("UNKNOWN");

  servoFeed->save(90);

}

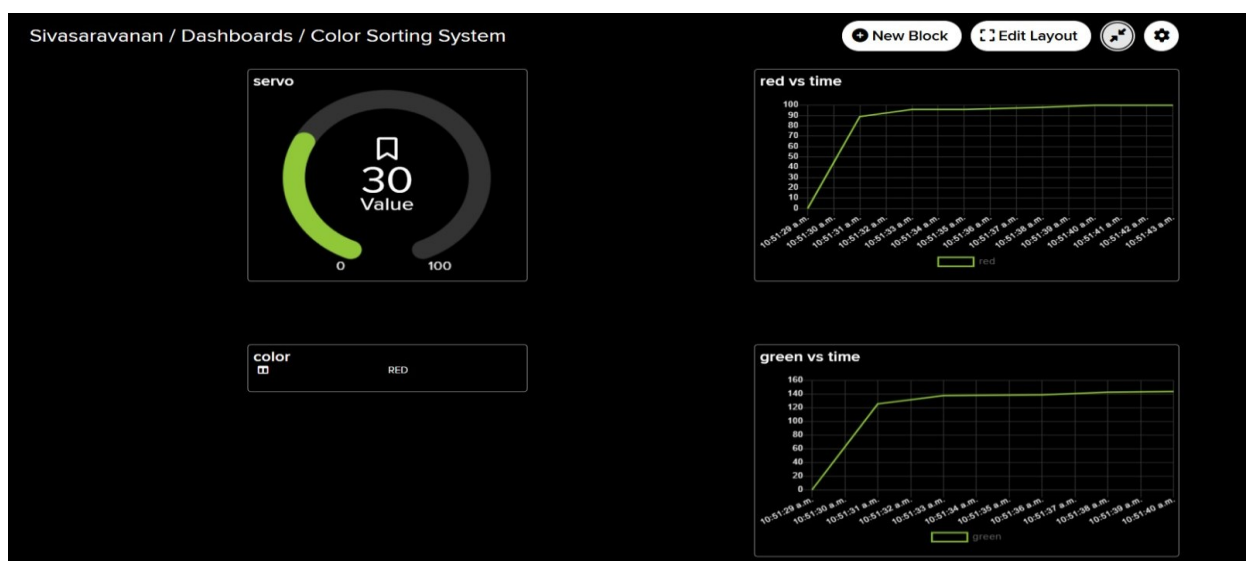
Serial.println("-----");

delay(1000);

}

```

9. System Working :



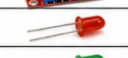
When the system is switched ON, the Arduino initializes all connected components. The conveyor motor begins rotating and moves products toward the sensing area.

When a product reaches the IR sensor, the sensor detects its presence and sends a signal to the Arduino. The Arduino temporarily stops the conveyor so that the product remains stationary during color measurement.

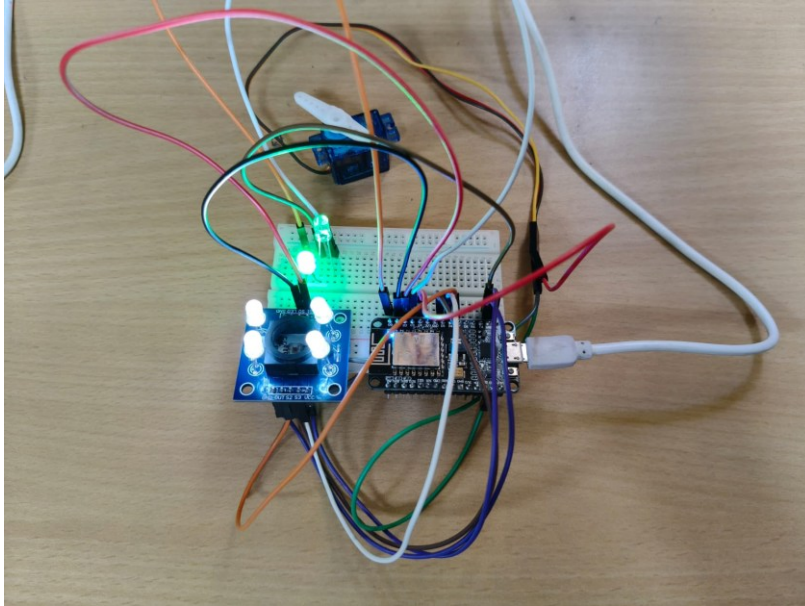
The TCS3200 sensor measures the reflected light from the product. The sensor generates frequency outputs corresponding to the red, green, and blue components of the detected color.

10. BILL OF MATERIAL:

BILL OF MATERIAL

S.No.	Component		Specification / Description	Quantity
1.		Arduino UNO	Arduino UNO R3 Development Board	1
2.		TCS3200 Color Sensor	TCS3200 Color Recognition Sensor Module	1
3.		IR Obstacle Sensor	IR Infrared Obstacle Detection Sensor Module	1
4.		Servo Motor	Standard Servo Motor (SG90) for Sorting Mechanism	1
5.		DC Gear Motor	300 RPM DC Gear Motor (for Conveyor Belt)	1
6.		L298N Motor Driver	L298N Dual H-Bridge Motor Driver Module	1
7.		Red LED	5mm LED (Red)	1
8.		Green LED	5mm LED (Green)	1
9.		Blue LED	5mm LED (Blue)	1
10.		Resistor	220 Ω Resistor (for LEDs)	3
11.		Breadboard	Solderless Breadboard	1
12.		Jumper Wires	Male to Male Jumper Wires	As required
13.		Power Supply	5V DC Power Supply for Arduino & Sensors (6V – 12V DC Power Supply for Motors)	2
14.		Conveyor Belt Mechanism	Mini Conveyor Belt for Product Movement	1

11. Prototype Implementation:



The prototype was constructed using an Arduino UNO as the main controller. A small conveyor mechanism was used to transport the products through the detection area.

The TCS3200 color sensor was mounted above the conveyor to identify the color of each product. An IR sensor was positioned near the detection area to identify when a product arrived.

The conveyor was operated using a DC motor controlled through an L298N motor driver. A servo motor was used as the sorting mechanism.

12. Results and Performance Analysis:

The Automated Color-Based Product Sorting System was successfully designed and implemented as a prototype for industrial automation applications. The system was tested to verify its ability to detect the presence of products, identify their colors, and automatically sort them into their respective categories.

During operation, the IR sensor successfully detected the arrival of a product on the conveyor belt. Once the product was detected, the conveyor

was stopped temporarily to provide a stable position for color sensing. The TCS3200 color sensor then measured the reflected light from the product and provided the corresponding color information to the Arduino UNO.

The Arduino UNO processed the sensor output and classified the product based on its dominant color. For the prototype, the system was designed to identify red, green, and blue products. After identifying the color, the corresponding LED was activated to provide a visual indication.

Performance Analysis:

The prototype performed successfully in the following areas

The prototype demonstrates that color-based product sorting can be automated using a microcontroller and color sensor. The system reduces the need for manual sorting and provides a consistent method for separating products.

The sorting performance depends on factors such as sensor distance, lighting conditions, product surface, conveyor speed, and color calibration. Proper calibration of the TCS3200 sensor improves the accuracy of color identification.

The system is suitable as a prototype for industrial automation applications. For large-scale industrial implementation, additional features such as high-speed conveyors, industrial-grade sensors, pneumatic actuators, machine vision, emergency-stop systems, and PLC control can be incorporated.

Overall, the prototype successfully demonstrates an automated color-based product detection and sorting system that can improve productivity and reduce human intervention in manufacturing processes.